SIMATS ENGINEERING
ASSESSMENT TOOL 2 - I

COURSE NAME: Cryptography and Network Security COURSE CODE: CSA5119

COURSE OUTCOME COVERED

CO1 : Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5 **Total Marks:50**

Annexure A

CONCEPT MAPPING

Code of Conduct:

I J. Vishwanth (192573030) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

J. Vishwanth

Signature of the student:

Annexure A

APPLICATION- BASED QUESTIONS

1. **Scenario: Event Communication Security**
A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.
   a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
   b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. **Scenario: Intercepted Suspicious Message**
A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.
   a. Deduce the most likely original message.
   b. Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

3. **Scenario: Secure Messaging System**
A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.
   a. Demonstrate how the encryption process transforms the message step-by-step.
   b. Explain how the repeating keyword influences each character of the ciphertext.

4. **Scenario: Suspicious Digital Asset Investigation**
During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.
   a. Analyze how investigators could detect the presence of hidden data within the image.
   b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. **Scenario: Legacy Encryption in Corporate Communication**
A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEEFEAOCAIVDEN".
   a. Determine the most probable original message by reversing the encryption process.
   b. Critically evaluate why such a method is vulnerable in modern communication systems.
   c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Steganography hides the existence of a secret message inside an ordinary carrier such as image or audio or ordinary.

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Caesar Cipher:-

(a) Shift each letter 4 places forward
(C = P+4 mod 26):

The scenario uses a Caesar cipher, a monoal substitution cipher. Every letter is replaced by a letter a fixed number of positions away in a alphabet. The shift value (here 4) is the secret key.

(a) Reconstructing the encrypted message (shift=4).

Number the alphabet A=0, B=1 ... Z=25

The encryption formula is

C=(P+K) K=4 C= Cyphertext position

Plaintext	M	E	E	T	A	T	N	O	O	N
P value	12	4	4	19	0	19	13	14	14	13
P+4	16	8	8	23	4	23	17	18	18	17
Cyphertext	Q	I	I	X	E	X	R	S	S	R

Spaces are not encrypted, so word boundaries stay visible

Encrypted message QIIX EXRSSR

Notice the both E is the HEET because, both T' became X, and both O became S. The same plaintext letter always give the same letter which is the main letter of the cipher

(b) An authorized member knows the cipher type (Caesar) and the key (shift 4). They apply the inverse operation

$P = (C - K) \bmod 26$

Steps

Write down the Ciphertext : QIIXEX RSSR.

Move letter 4 position backward in alphabet

If the shift goes before A, it wrap around to Z (this is the mod 26 part).

Cipher	Q	I	I	X	E	X	R	S	S	R
C − 4	12	4	4	19	0	19	13	14	14	15
Plain	M	E	E	T	A	T	N	O	O	N

4. Steganography hides the existence of a secret message inside an ordinary carrier such as image audio (or) ordinary carrier such as image, audio video file. Unlike cryptography which makes a message unreadable the message unnoticable & common method is LSB. The last bit of pixel value. This changes the colour so lightly that eye cannot see it.

(a) Compare with original file

Compare file size, dimension and format with a known clean copy of the logo

MD5 (SHA-256) - 1Bit.

Large size with identical appearance, strong sign of embedded data

(b) Encrypt and compress the payload before embedding. This hidden bit looks random and unreadable even if extracted

(b) Adaptive (or) transform domain

Hide data in noisy (or) textured regions with a k based rand pixel order. This reduces visible and d do its LSB and Chi-square detection